

BUILDMESH

Construction intelligence

that observes, reasons, and keeps people in control.

A local-first project graph and OpenMesh agent runtime for evidence-backed construction workflows.

Qualcomm Snapdragon AI Lab Build & Present Challenge 2026

Backend MVP + deployment architecture

Construction evidence is fragmented; decisions are delayed.

Reality is scattered

Photos, worker reports, drawings, schedules, materials, weather, and decisions live in separate places. Teams manually reconcile them.

AI must be accountable

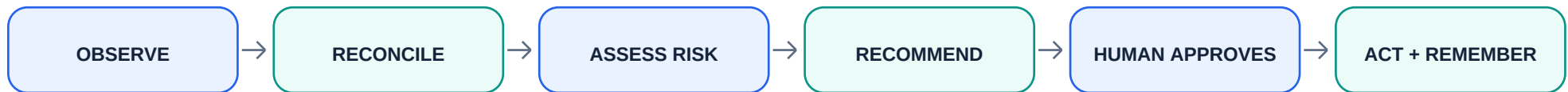
A generic chatbot cannot safely change a construction plan. Findings need source evidence, confidence, review, and a durable decision record.

Cloud is not the default

Site images, drawings, and project context can be sensitive. Perception and project memory should remain on the device whenever possible.

BuildMesh is not a promise to automate engineering judgment. It is the missing intelligence layer that turns project signals into reviewable, traceable work.

One evidence-backed loop, end to end.



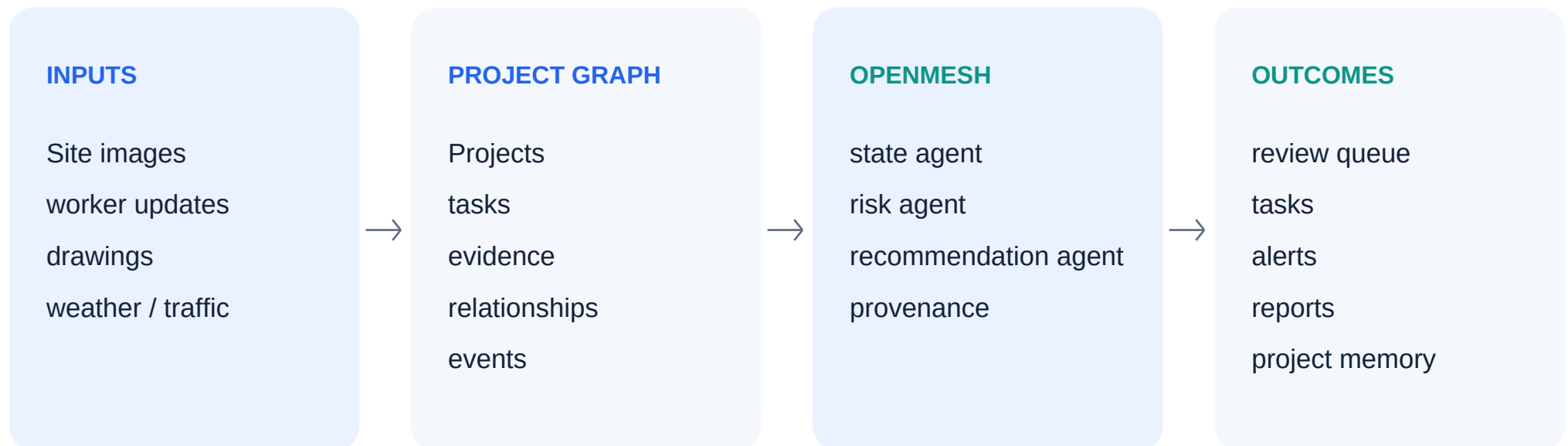
Example: Foundation F-12

A worker reports 85% completion and 35.5 m3 placed against a 42 m3 plan. BuildMesh retains the report as evidence, derives progress, checks consistency, and joins a sourced rain forecast.

Recommendation (pending site engineer review)

Review weather-sensitive waterproofing before forecast rain. The follow-up task does not exist until a named human approves it.

BuildMesh is the domain layer. OpenMesh is the orchestration fabric.



Every agent run stores input and output hashes. Every recommendation carries evidence IDs. Every consequential action is held behind an approval gate.

Local AI where privacy and latency matter; APIs only for inherently external data.

On the Snapdragon PC

Camera and image inference, document extraction, project graph, agent runtime, evidence store, local report generation. A QNN/ONNX command adapter isolates the device inference implementation.

External by nature

Weather, traffic, maps, and public environmental data remain sourced external context. Their provider, timestamp, payload, and confidence are recorded alongside the recommendation.

Honest validation plan

Benchmark the selected Qualcomm AI Hub or open-source model on the target HP Snapdragon device. Publish model version, image size, provider, end-to-end latency, power method, network state, and repeatable test protocol. No benchmark figure is claimed before measurement.

The deliverable proves the product thesis without pretending to solve all of construction.

v0.1 - now

Persistent SQLite project graph, API, evidence ingestion, state reconciliation, weather risk, recommendations, approval gate, task materialization, provenance, and tests.

Next integration

QNN-backed site vision, file storage, CAD/BIM ingestion, report rendering, notifications, role-based access, and target-device benchmarking.

Long-term thesis

BuildMesh scales from construction intelligence to infrastructure and built-environment intelligence while reusing the OpenMesh runtime.

BuildMesh makes the physical project legible to people - before it tries to make it autonomous.

Demo: observe -> understand -> recommend -> human approval -> action -> project memory